

Current Mode Logic Design and Simulation

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ADI Internship

Assignment: Current Mode Logic Design and Simulation

Objective

This tutorial explores the tradeoffs of CML (Current Mode Logic) design through a CML inverter test bench: DC biasing, swept-DC transfer characteristics, small-signal frequency response (gain, CMRR, PSRR), and large-signal transient/PSS+PNOISE behavior (delay, slew rate, and jitter) across two edge-rate/frequency corners.

Part 1: CML Inverter — Design Conditions

- $V_{dd} = 1.5 \text{ V}$, $V_{swing} = 300 \text{ mV}$ (differential output peak, set as design variable)
- $V_{in,CM} = V_{dd} - V_{swing} = 1.2 \text{ V}$ (kept smaller than $V_{out,CM}$)
- Differential pairs: $L = L_{min}$, $I_D/W = 70 \text{ } \mu\text{A}/\mu\text{m}$
- Tail source: $L > 1 \text{ } \mu\text{m}$, $I_{ss}/W = 10 \text{ } \mu\text{A}/\mu\text{m}$
- $R_D = 300 \text{ } \Omega$, $C_L = 100 \text{ fF}$
- Tail current set as a function of V_{swing} and R_D : $I_{ss} = V_{swing} / R_D$

1. Schematic

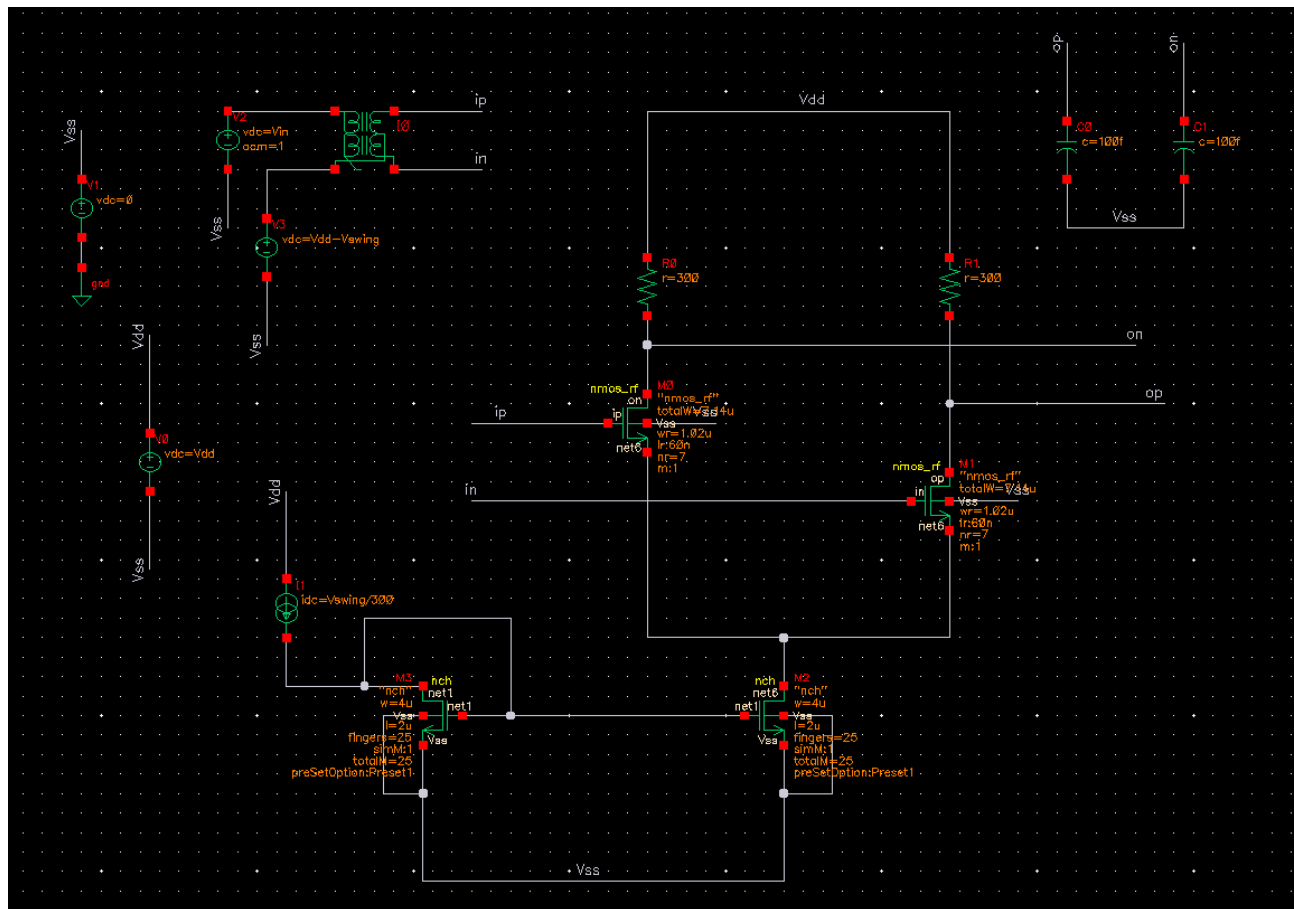


Figure 1 — CML buffer Test Bench — $V_{dd} = 1.5\text{ V}$, $V_{swing} = 300\text{ mV}$, $R_D = 300\ \Omega$, $C_L = 100\text{ fF}$, differential pairs $L = L_{min}$ ($ID/W = 70\ \mu\text{A}/\mu\text{m}$), tail $L > 1\ \mu\text{m}$ ($I_{ss}/W = 10\ \mu\text{A}/\mu\text{m}$)

2. DC Operating Point

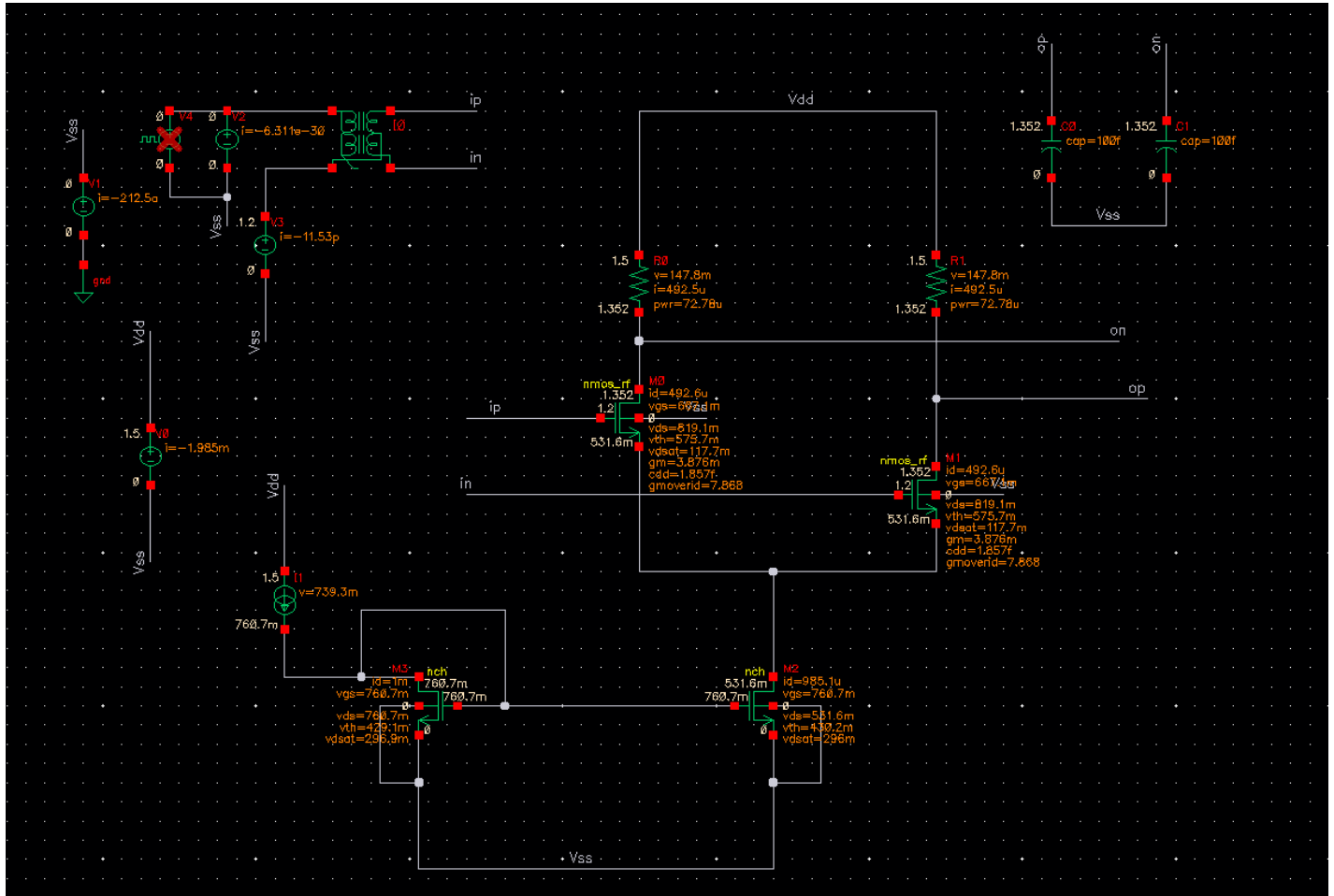


Figure 2 — Annotated DC Operating Point — $V_{in} = 0$ (differential input at common-mode, $V_{in,CM} = 1.2V$)

Take note of the VOV and VTH of the input pair (why):

- VOV — needed to ensure full current steering of I_{ss} between the two branches over the input swing range.
- VTH — sets the achievable output common-mode / headroom and, together with VOV, sets the maximum available differential swing.

Check that all MOSFETs are in saturation: verified for all devices, $V_{DS} > V_{DSat}$ in the annotated operating point above.

3. Swept DC Analysis

V_{in} swept from $-2 \cdot V_{swing}$ to $+2 \cdot V_{swing}$ (-0.6 V to $+0.6$ V).

Differential and Single-Ended Output Voltage

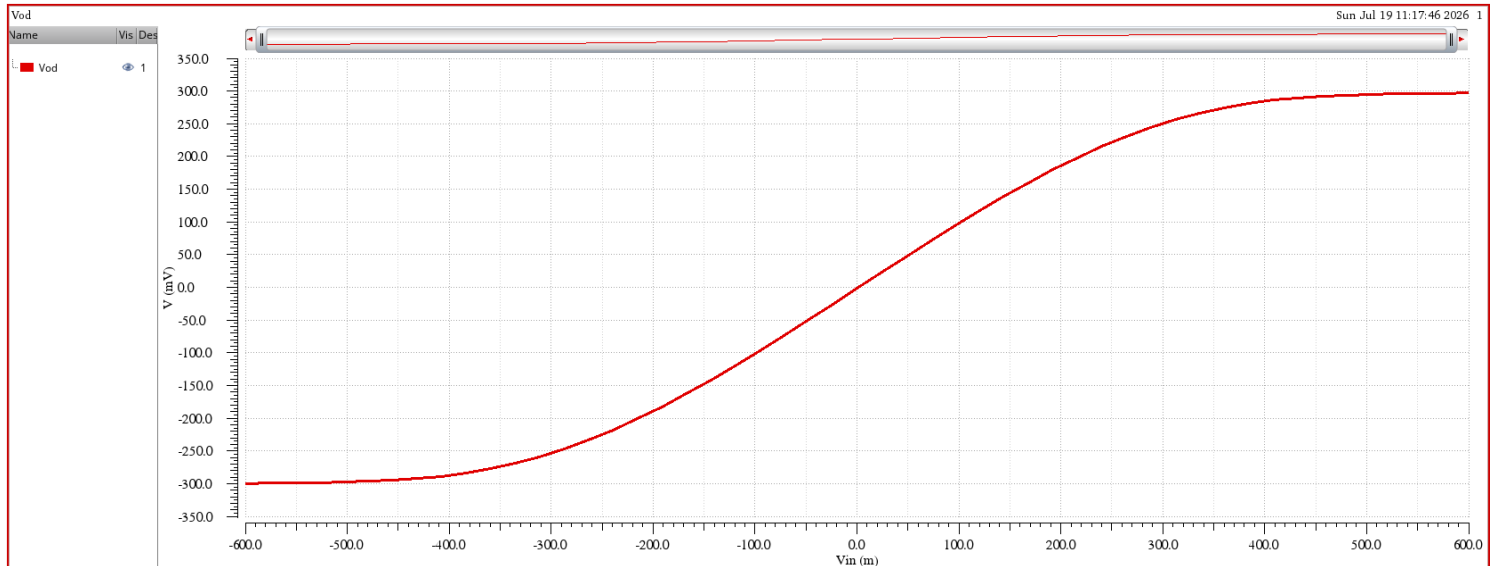


Figure 3 — Differential Output Voltage vs. V_{in} — $V_{in}: -2V_{swing} \rightarrow +2V_{swing}$

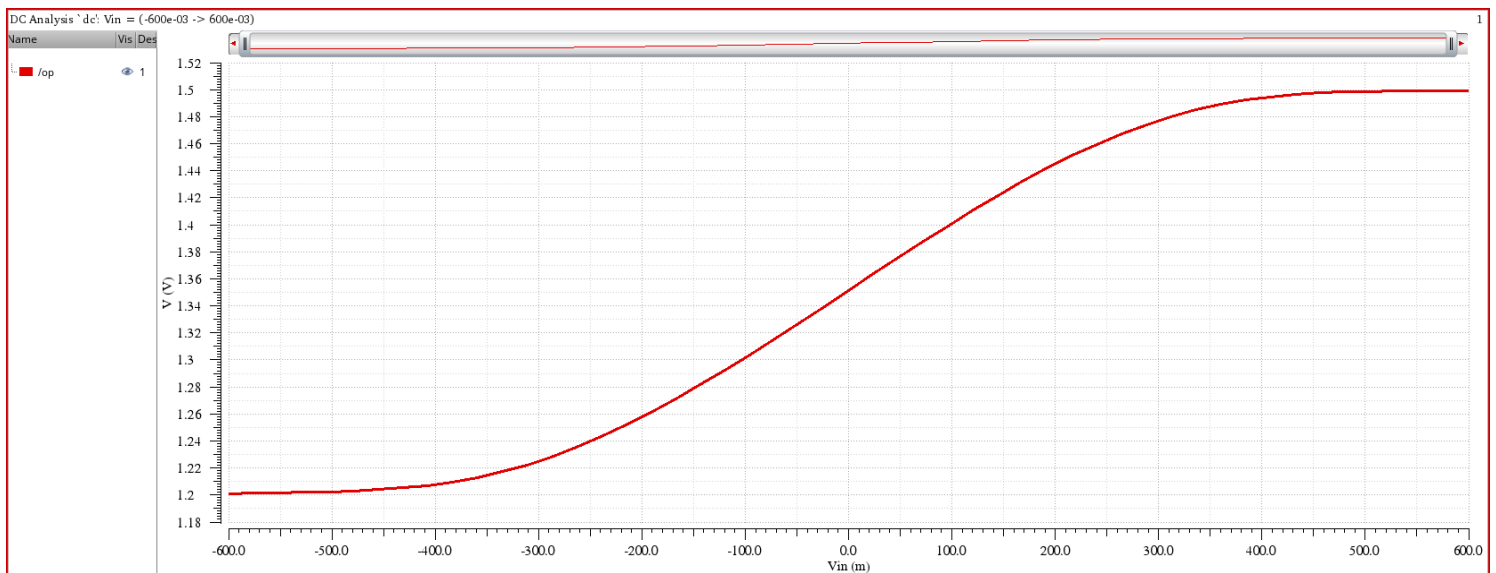


Figure 4 — Single-Ended Output Voltage vs. V_{in} — $V_{in}: -2V_{swing} \rightarrow +2V_{swing}$

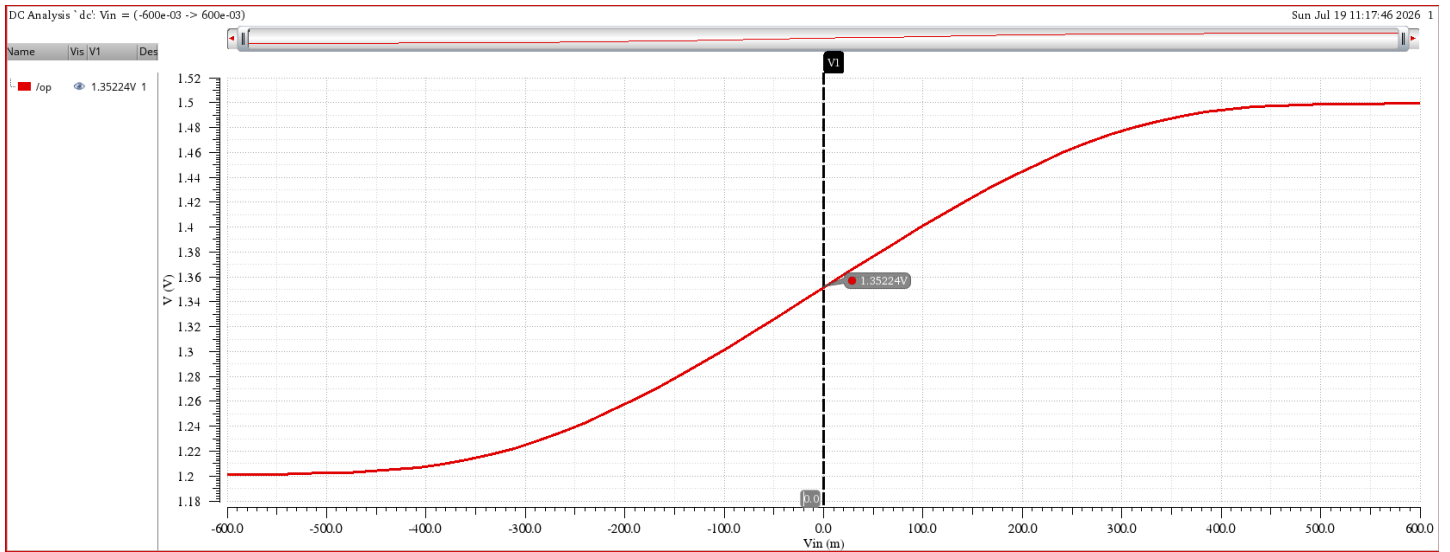


Figure 5 — Output Swing / Common-Mode Verification — $V_{in}: -2V_{swing} \rightarrow +2V_{swing}$

Comment: $Diff.swing = I_{ss} \cdot R_D = 0.3$ V; $Output\ CM = V_{dd} - \left(\frac{I_{ss}}{2}\right) \cdot R_D = 1.35$ V — both match the design targets.

Differential Branch Current

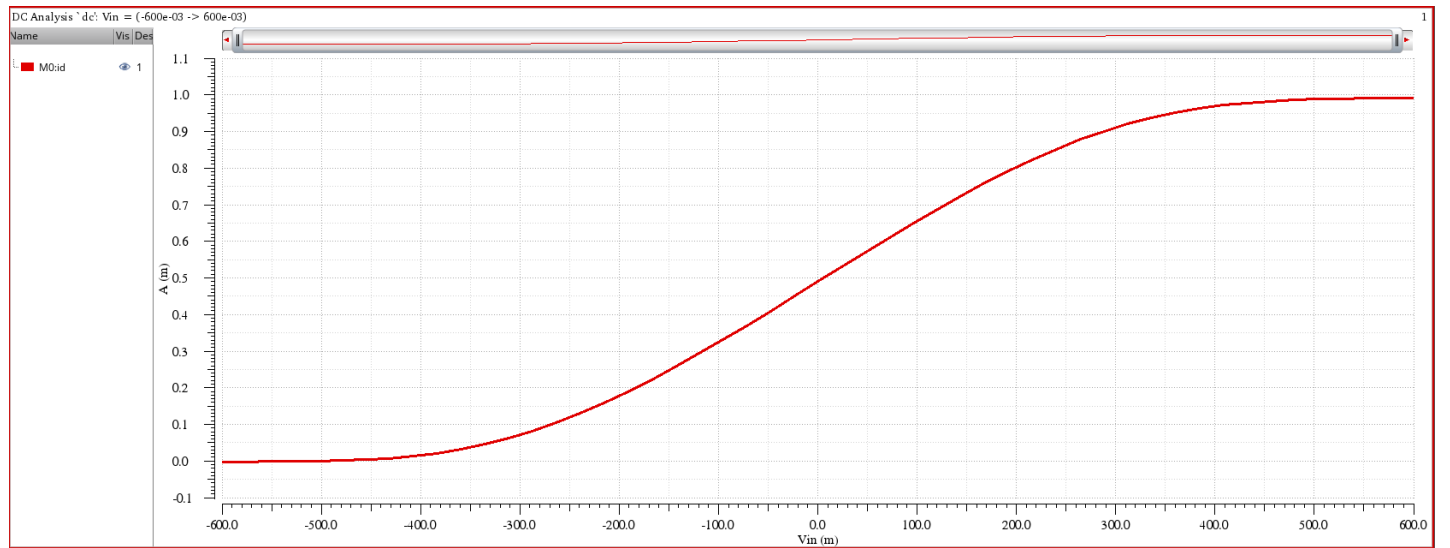


Figure 6 — Current in One Differential Branch vs. V_{in} — $V_{in}: -2V_{swing} \rightarrow +2V_{swing}$

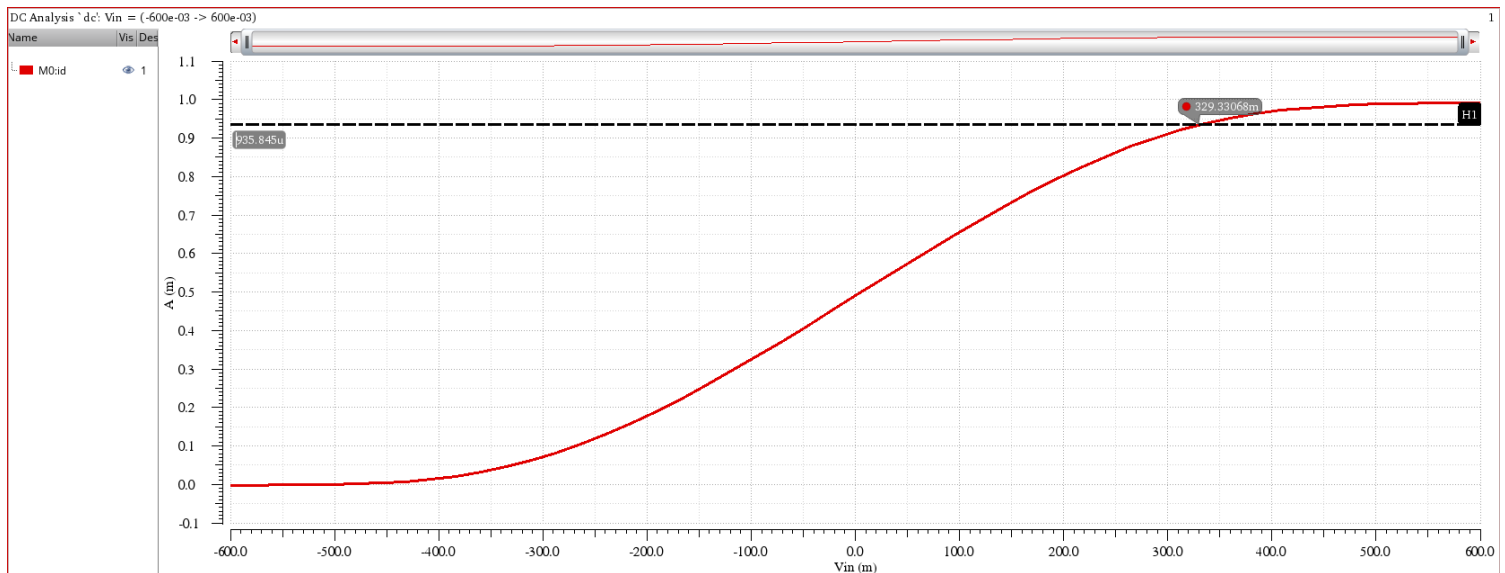


Figure 7 — 95% Current Steering Point — V_{in} : $-2V_{swing} \rightarrow +2V_{swing}$
Comment: 95% of I_{ss} flows in one branch at a differential input of 329.33 mV.

Virtual Ground Node VP

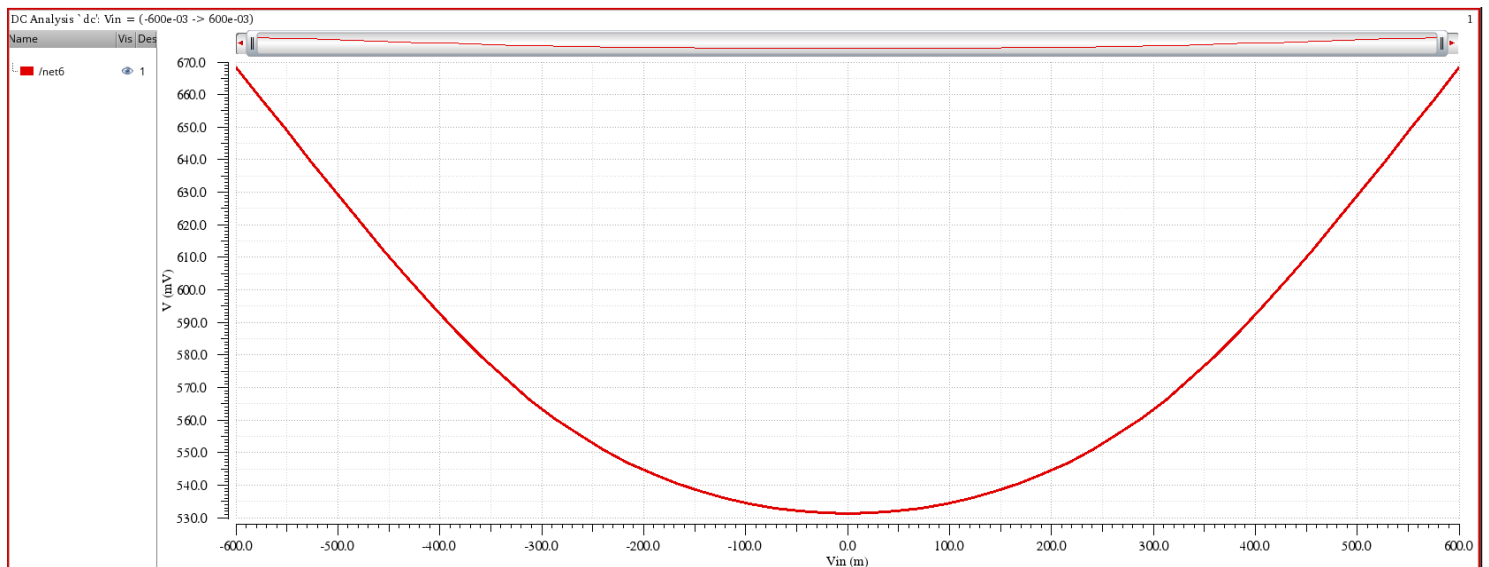


Figure 8 — Voltage at Virtual Ground Node VP vs. V_{in} — V_{in} : $-2V_{swing} \rightarrow +2V_{swing}$
Comment: VP shows only a small swing — the tail device keeps I_{ss} essentially constant, so the virtual ground barely moves as the input pair steers current between the two branches.

4. XF (Transfer Function) Analysis

Frequency swept 1 GHz → 10 GHz; differential, common-mode, and supply XF source magnitudes set to 1.

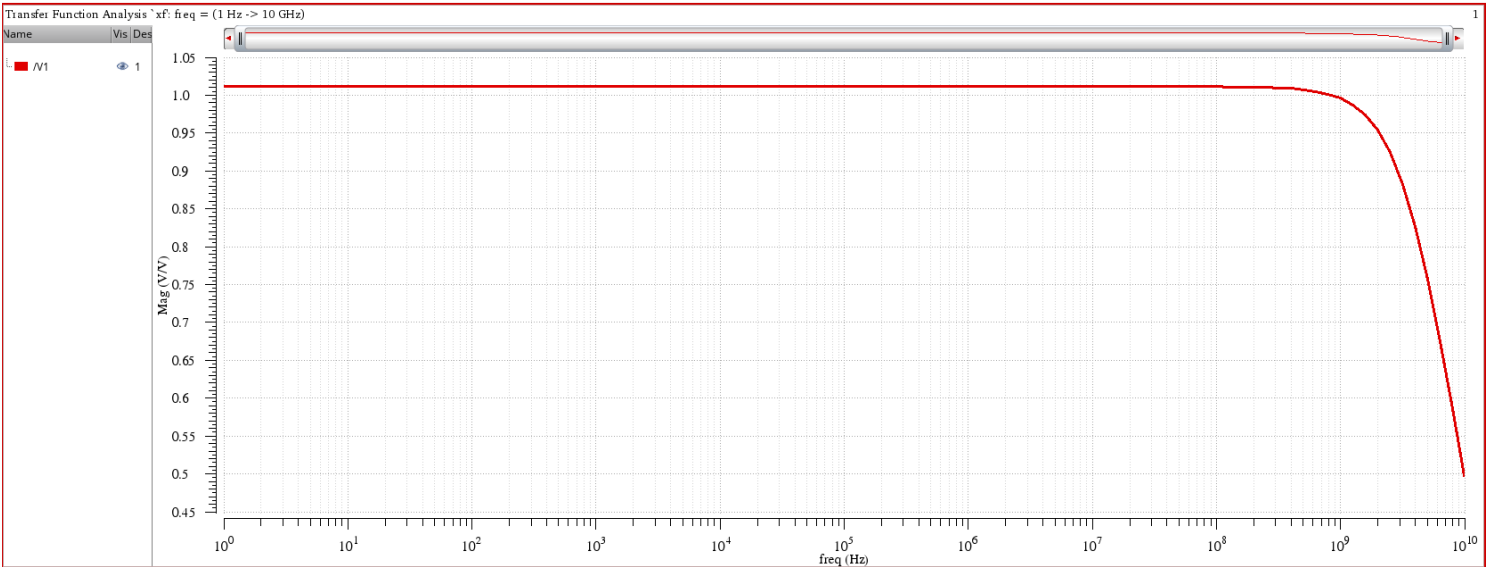


Figure 9 — Differential Gain Avd vs. Frequency — 1 GHz → 10 GHz, diff. XF mag = 1

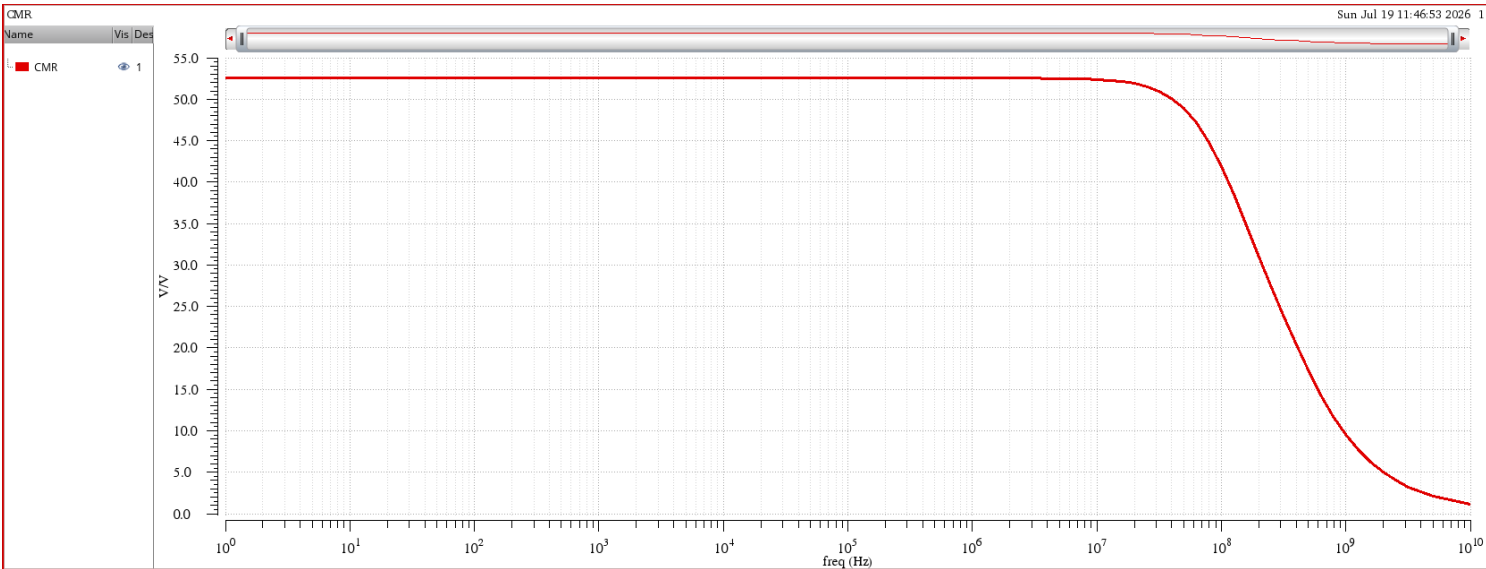


Figure 10 — Single-Ended CMRR (linear) vs. Frequency — 1 GHz → 10 GHz, CM XF mag = 1

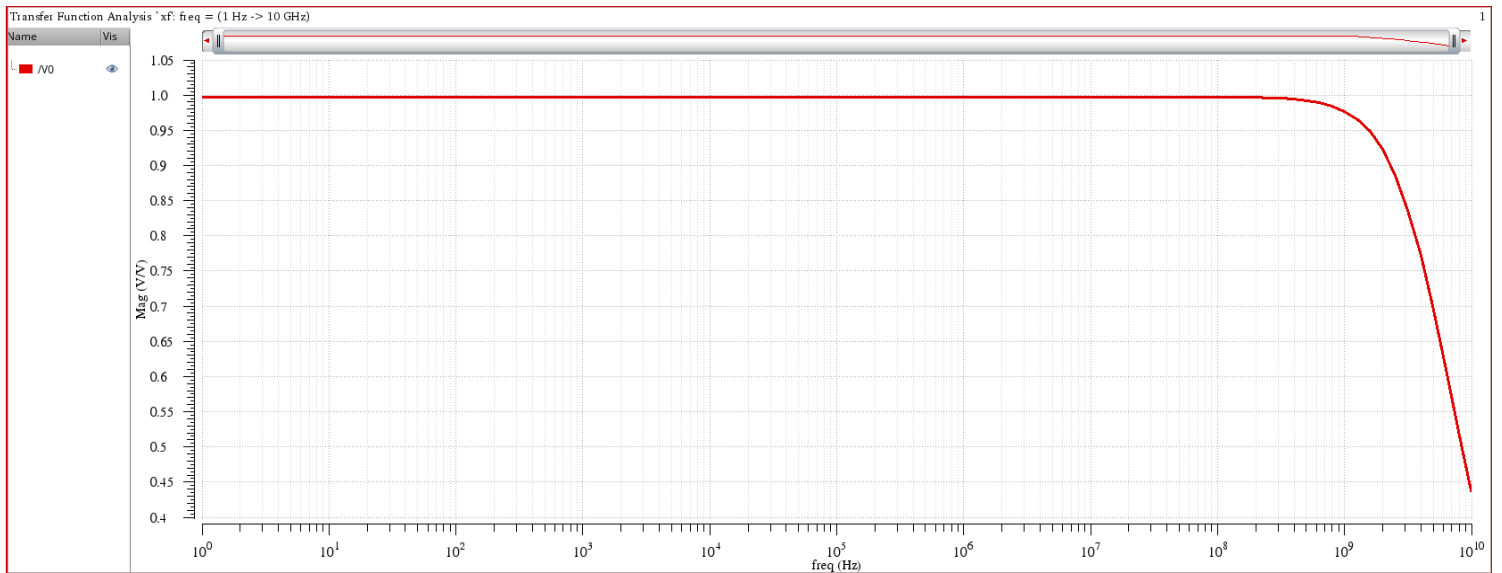


Figure 11 — Single-Ended PSRR (linear) vs. Frequency — 1 GHz $\rightarrow$ 10 GHz, supply XF mag = 1

5. PSS + PNOISE — frq = 1 GHz, tfr = 20 ps

Differential input source replaced with a vpulse: V1 = $-V_{in}$, V2 = $+V_{in}$, period = $1/\text{frq}$, delay = $0.5/\text{frq}$, rise = fall = $\text{tfr} \cdot V_{in}/V_{\text{swing}}$ (constant slew rate at zero-crossing across the swing).

Input/Output Waveforms

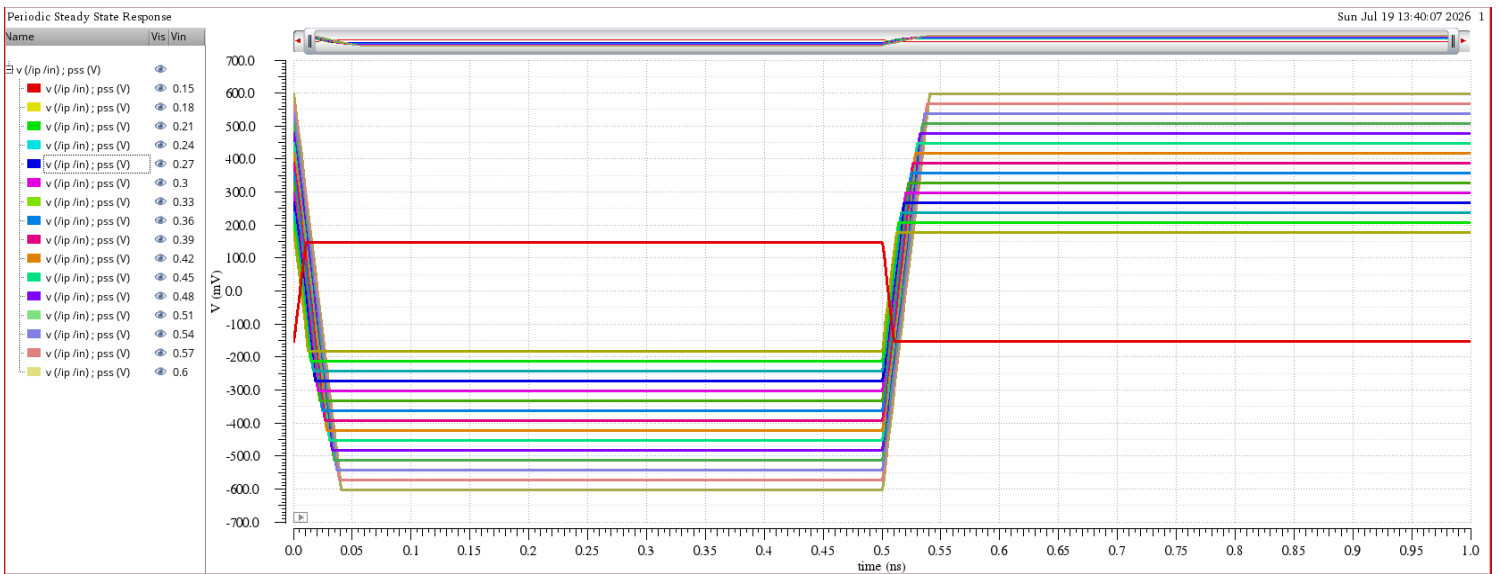


Figure 12 — Differential Input Waveform vs. Time — frq = 1 GHz, tfr = 20 ps, swept over V_{in}

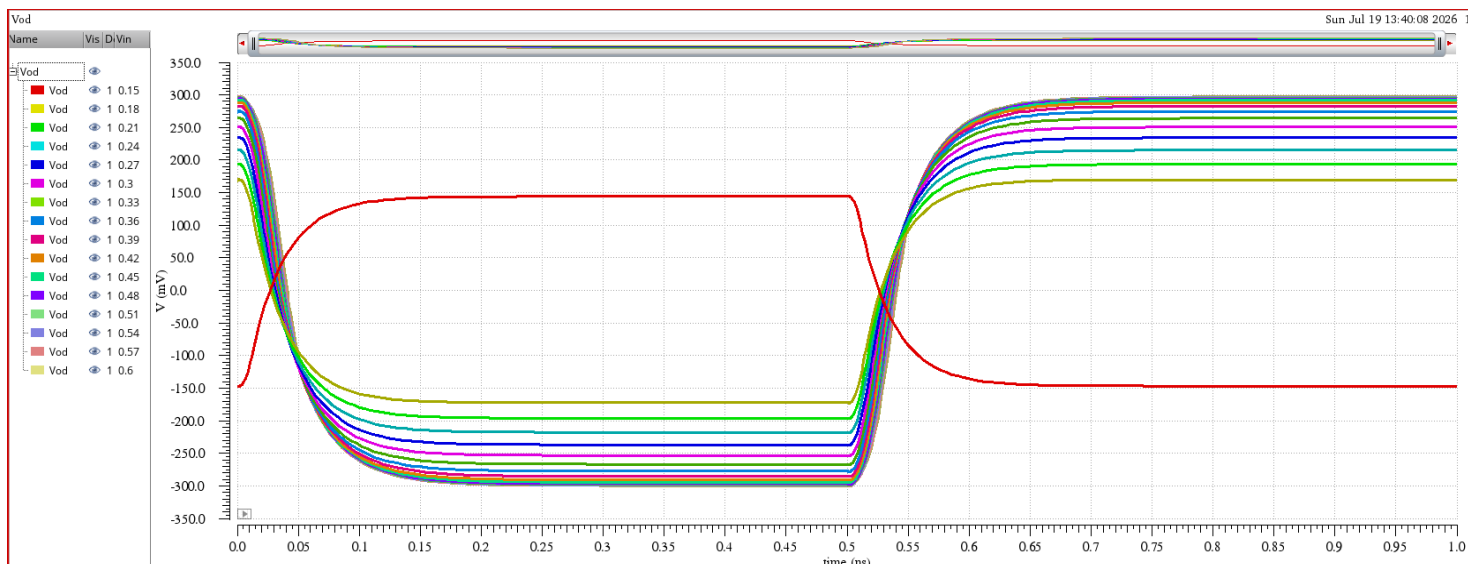


Figure 13 — Differential Output Waveform vs. Time — $f_{rq} = 1 \text{ GHz}$, $t_{fr} = 20 \text{ ps}$, swept over V_{in}

Swing, Rise Time, Delay

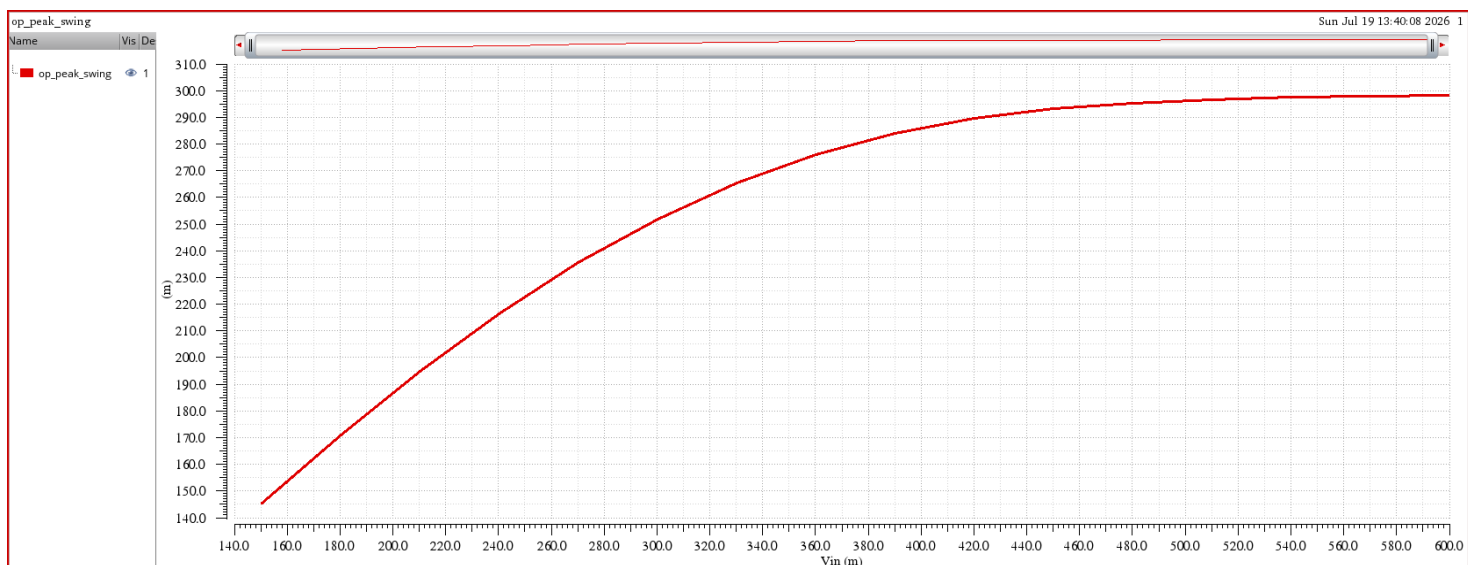


Figure 14 — Differential Output Peak Swing vs. V_{in} — $f_{rq} = 1 \text{ GHz}$, $t_{fr} = 20 \text{ ps}$

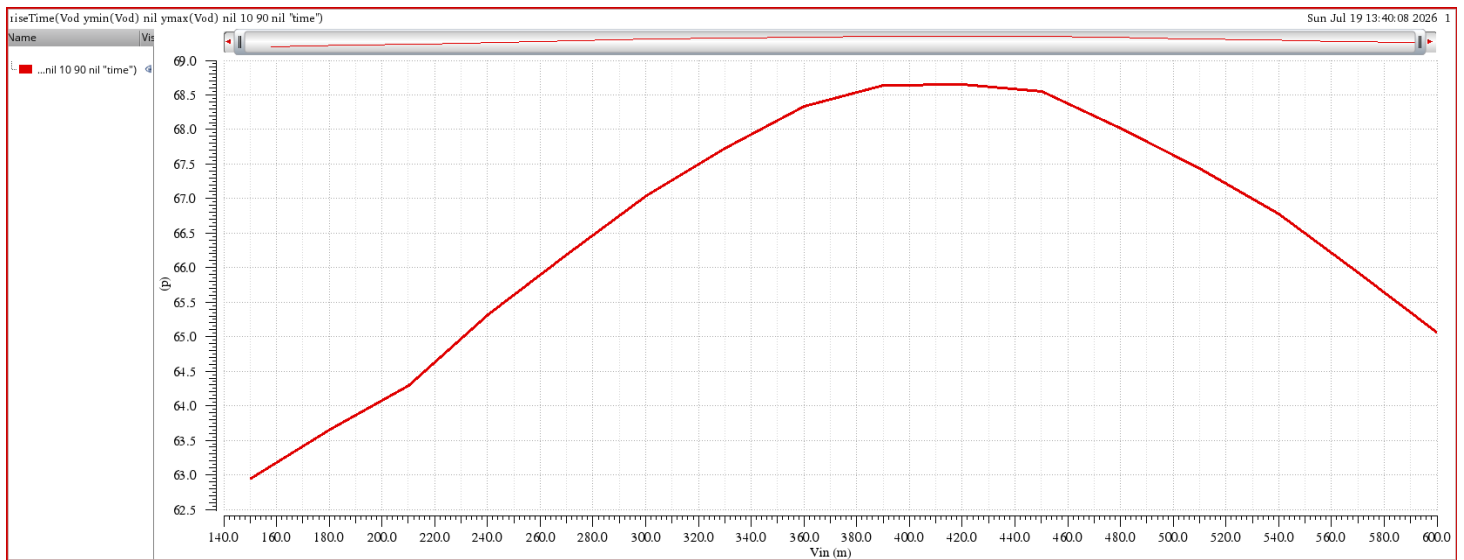


Figure 15 — Output Rise Time vs. Vin — $f_{rq} = 1 \text{ GHz}$, $t_{fr} = 20 \text{ ps}$



Figure 16 — Low-to-High Input-to-Output Delay vs. Vin — $f_{rq} = 1 \text{ GHz}$, $t_{fr} = 20 \text{ ps}$

Comment: Hand calc: $\text{delay} \approx 0.69 \cdot (C_L + C_{dd}) \cdot R_D = 0.69 \cdot (100 \text{ fF} + 1.85 \text{ fF}) \cdot 300 \Omega \approx 21.08 \text{ ps}$. This is very close to the simulated value — the small deviation is expected since the hand formula is a first-order RC approximation and neglects higher-order/nonlinear effects.

Slew Rate and Jitter

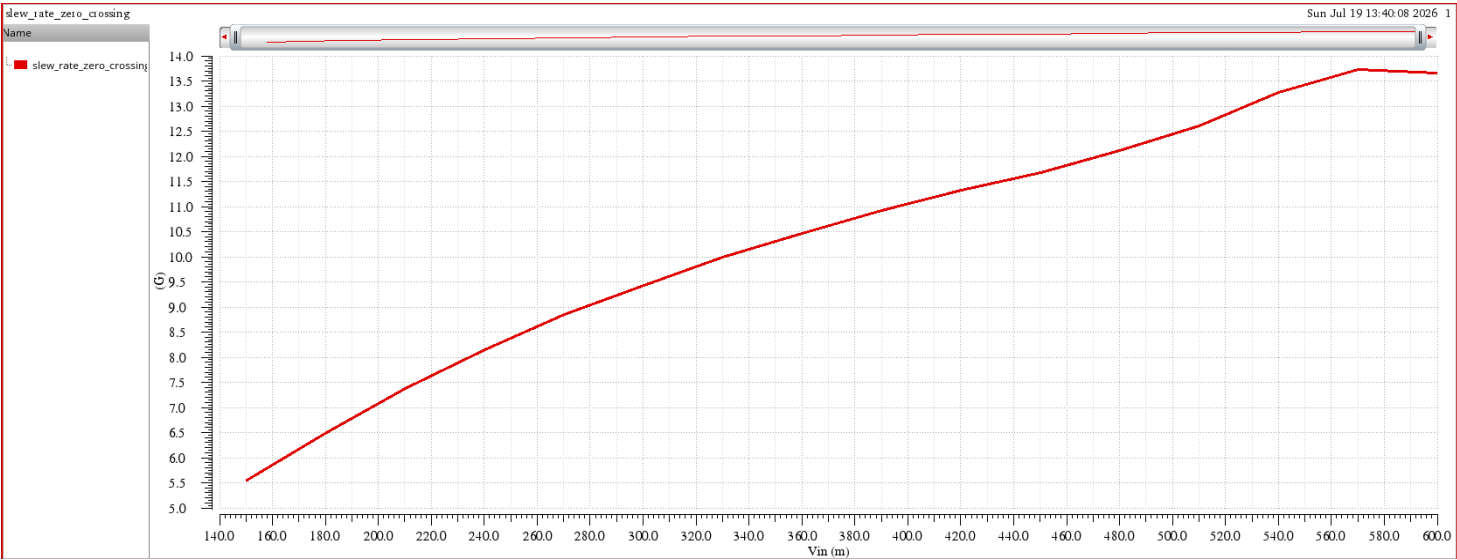


Figure 17 — Output Slew Rate at Zero-Crossing vs. Vin — $f_{rq} = 1\text{ GHz}$, $t_{fr} = 20\text{ ps}$

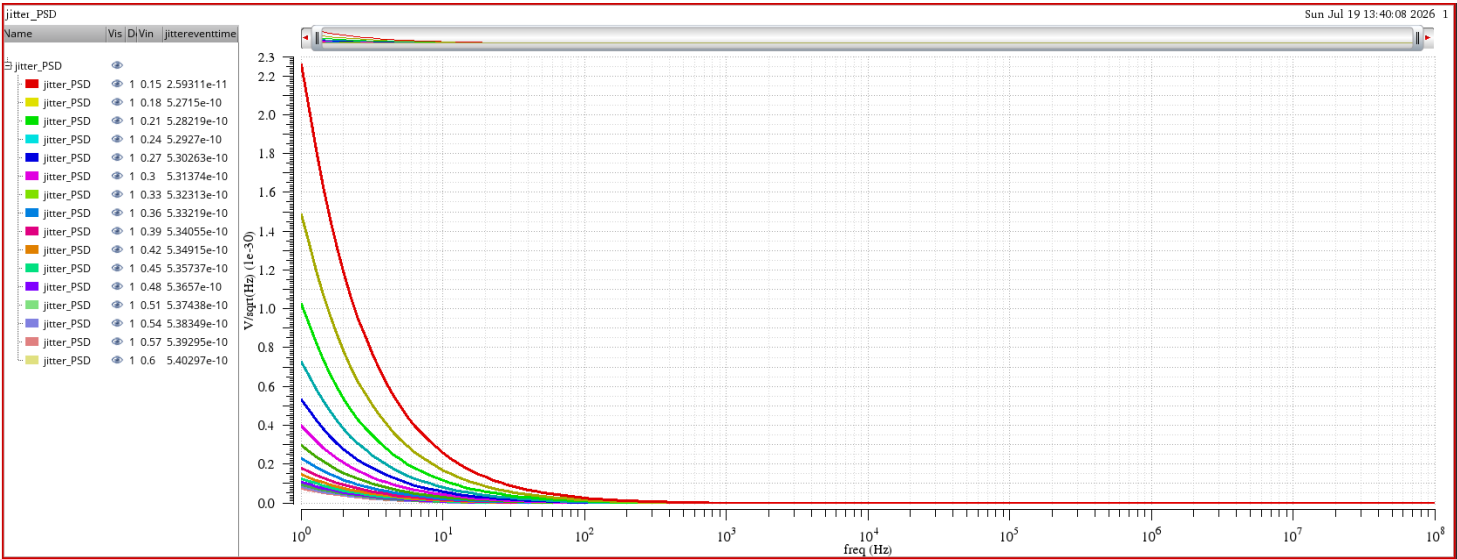


Figure 18 — Output Jitter PSD vs. Frequency — $f_{rq} = 1\text{ GHz}$, $t_{fr} = 20\text{ ps}$, swept over V_{in}

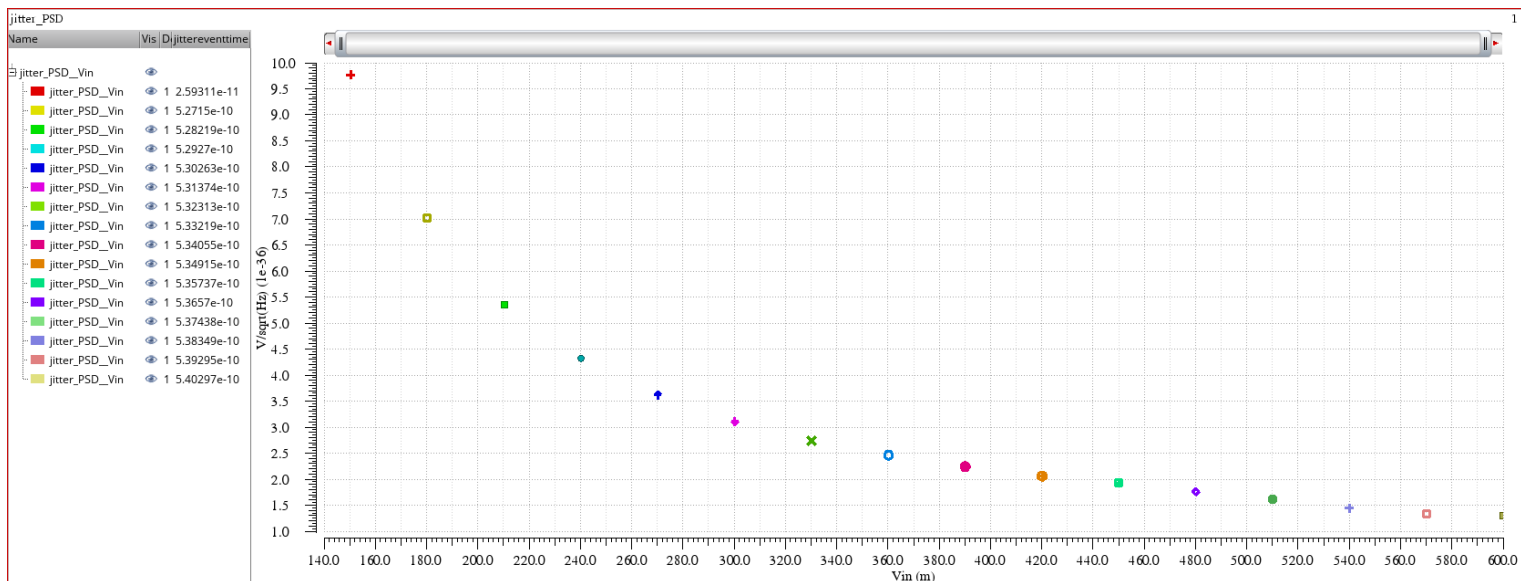


Figure 19 — Output Jitter at 10 MHz vs. Vin — $frq = 1\text{ GHz}$, $tfr = 20\text{ ps}$

6. Repeat: $frq = 5\text{ GHz}$, $tfr = 2\text{ ps}$

Input/Output Waveforms

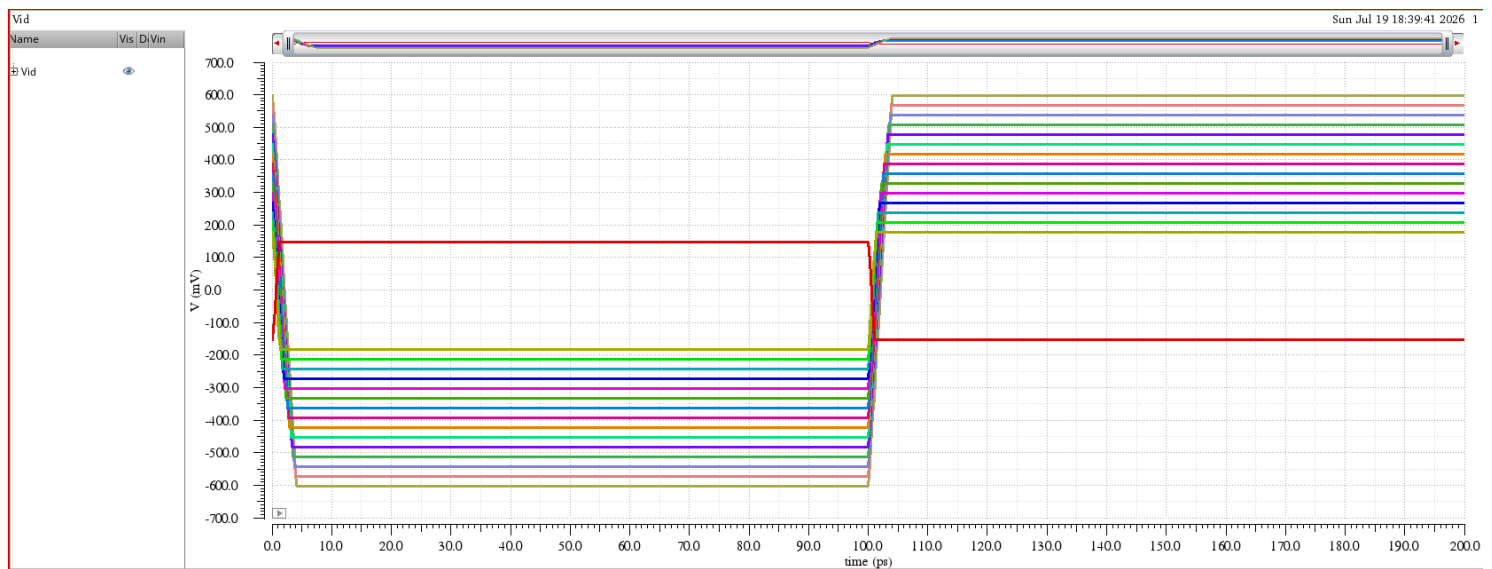


Figure 20 — Differential Input Waveform vs. Time — $frq = 5\text{ GHz}$, $tfr = 2\text{ ps}$, swept over Vin

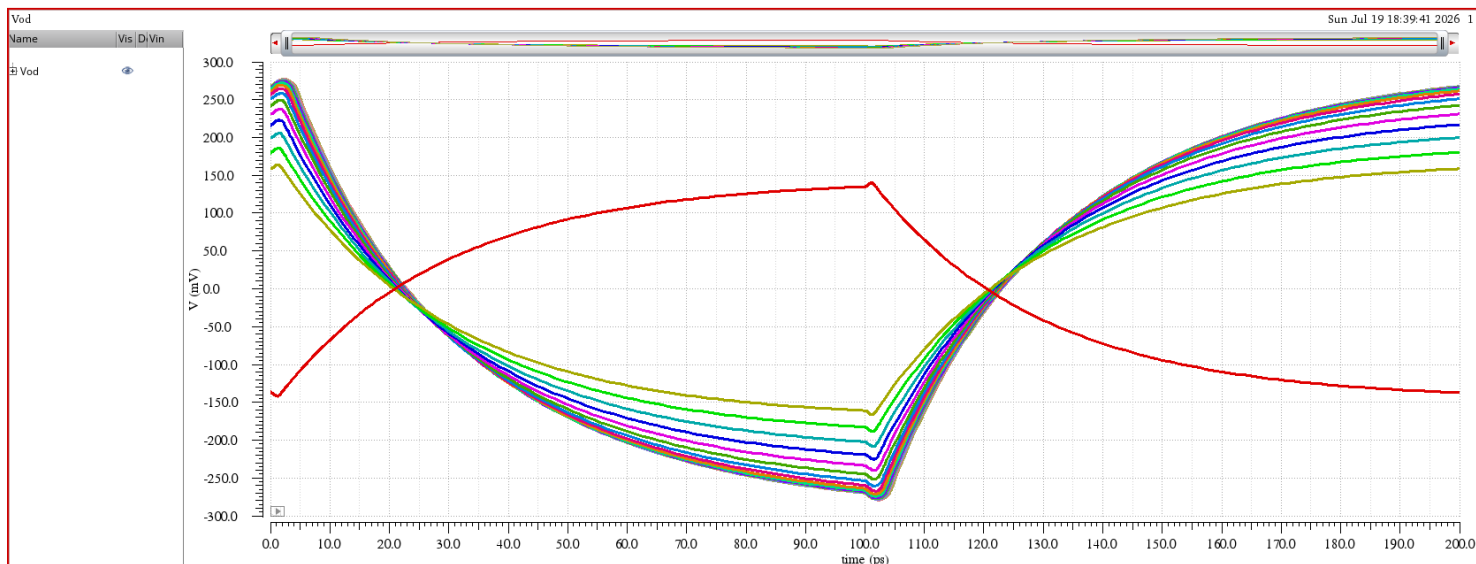


Figure 21 — Differential Output Waveform vs. Time — $frq = 5\text{ GHz}$, $tfr = 2\text{ ps}$, swept over V_{in}

Swing, Rise Time, Delay

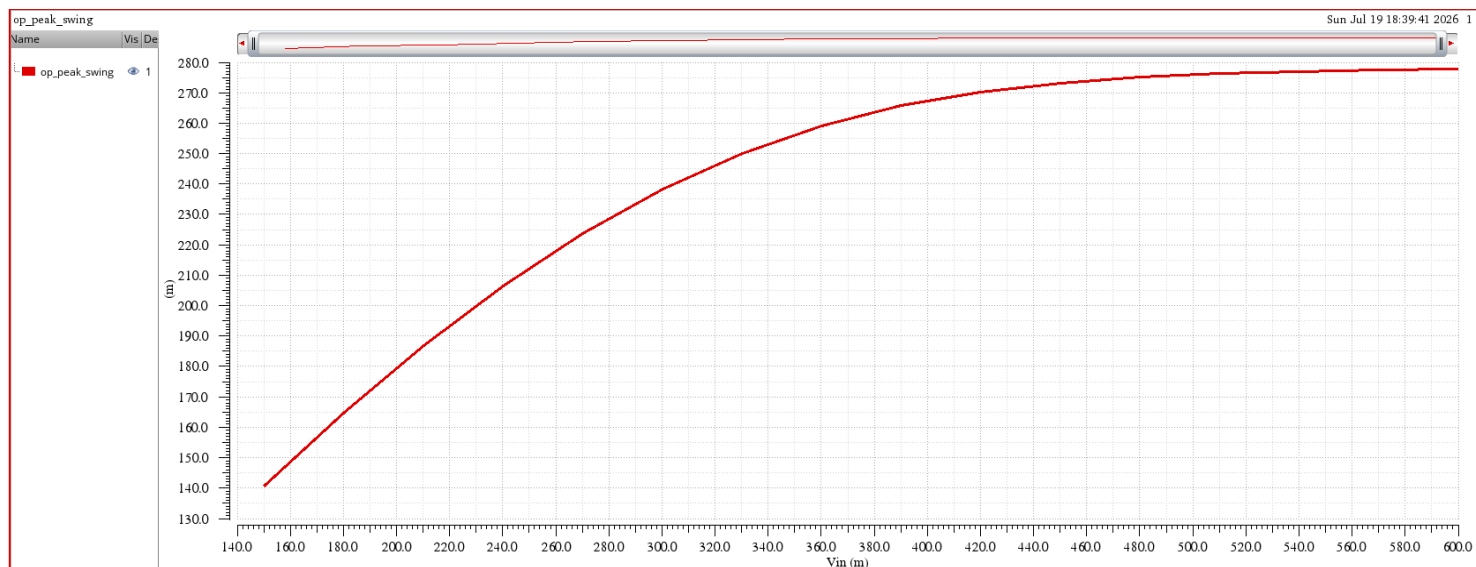


Figure 22 — Differential Output Peak Swing vs. Vin — $frq = 5\text{ GHz}$, $tfr = 2\text{ ps}$

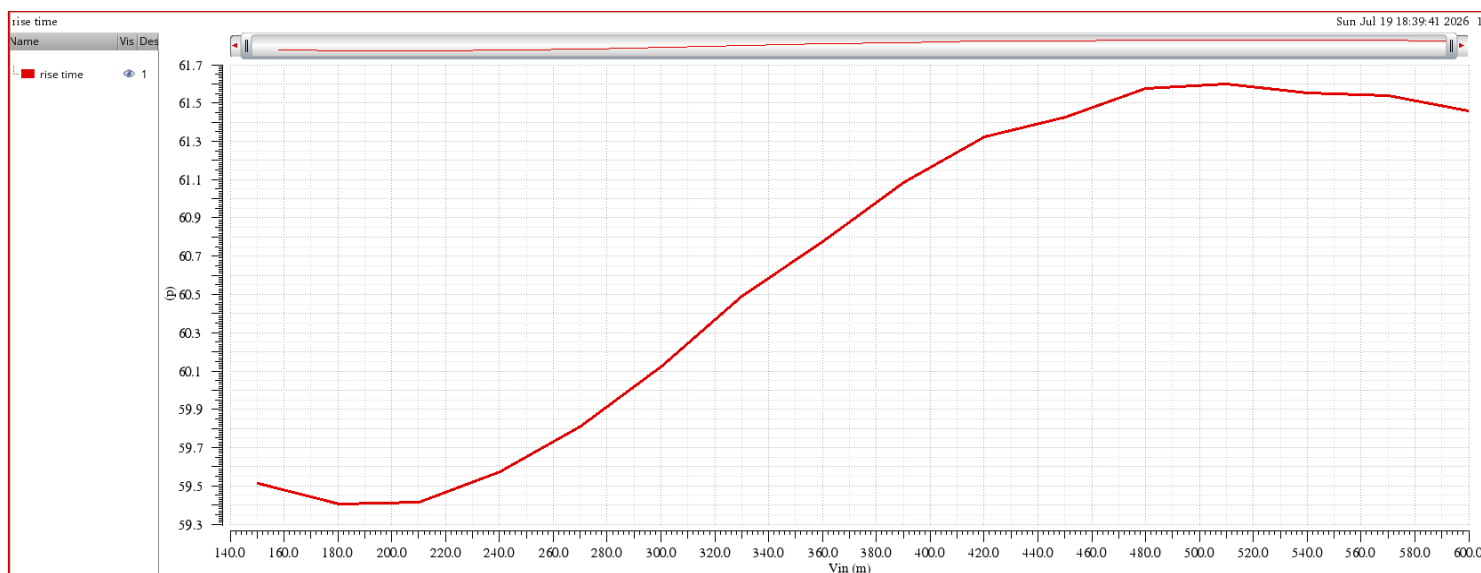


Figure 23 — Output Rise Time vs. V_{in} — $frq = 5\text{ GHz}$, $tfr = 2\text{ ps}$

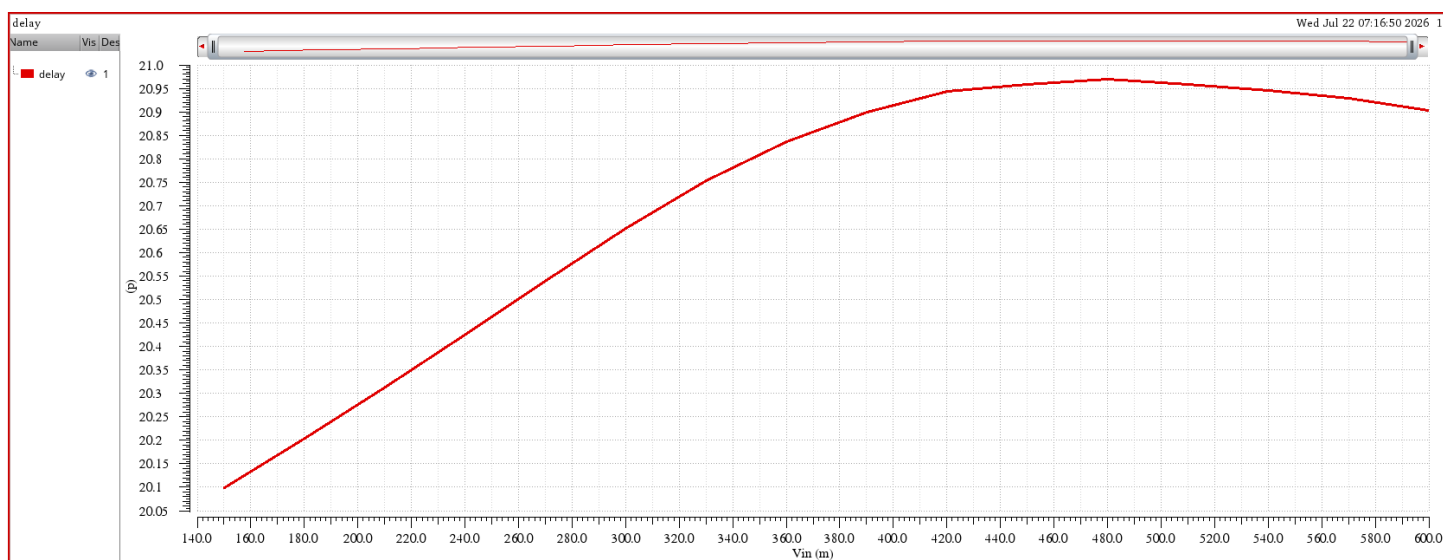


Figure 24 — Low-to-High Input-to-Output Delay vs. V_{in} — $frq = 5\text{ GHz}$, $tfr = 2\text{ ps}$

Slew Rate and Jitter

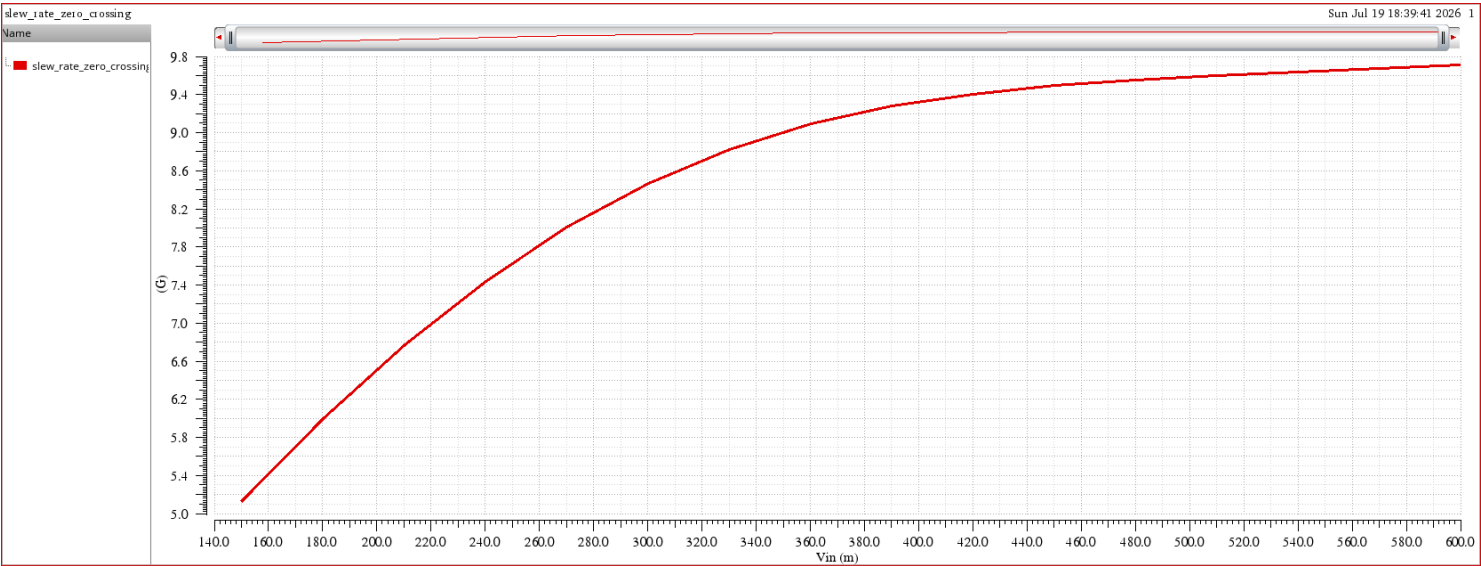


Figure 25 — Output Slew Rate at Zero-Crossing vs. Vin — $f_{rq} = 5\text{ GHz}$, $t_{fr} = 2\text{ ps}$

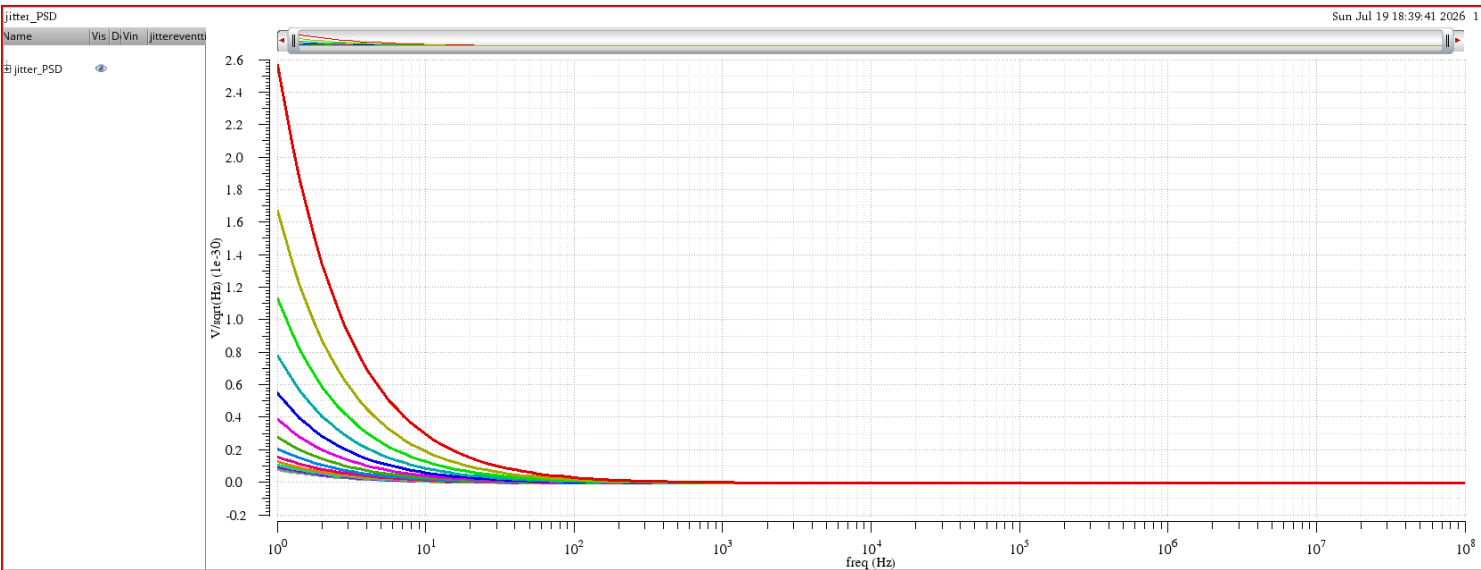


Figure 26 — Output Jitter PSD vs. Frequency — $f_{rq} = 5\text{ GHz}$, $t_{fr} = 2\text{ ps}$, swept over V_{in}

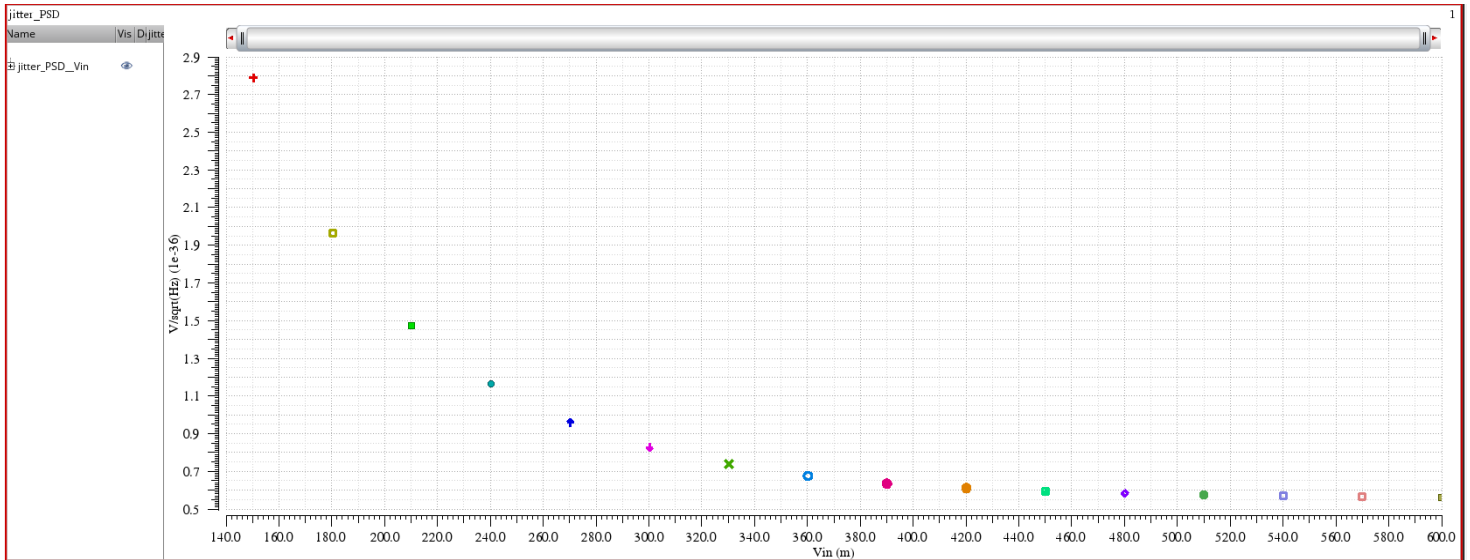


Figure 27 — Output Jitter at 10 MHz vs. Vin — $frq = 5\text{ GHz}$, $tfr = 2\text{ ps}$

Did the delay, output slew rate, or output swing change? Why?

Yes — all three changed relative to the 1 GHz / 20 ps corner:

- **Delay is almost the same** .
- **Slew rate decreases** — the output is still limited by the same RD·CL time constant; it cannot slew faster just because the input edge is faster, so relative to the shorter 5 GHz period the effective zero-crossing slope observed per cycle is reduced.
- **Output swing decreases** — at 5 GHz the bit period (200 ps) approaches the same order as the RC settling time, so the output no longer has enough time to fully settle before the next transition begins, clipping the peak-to-peak swing below the DC value of $I_{ss} \cdot RD$.

Appendix: Expressions Used

Test	Name	Type	Details	EvalType
Assignments:CML:1	Vod_dc	expr	$(v("/op" ?result "dc") - v("/on" ?result "dc"))$	point
Assignments:CML:1	CMR	expr	$(mag(getData("/V2" ?result "xf")) / mag(getData("/V3" ?result "xf")))$	point
Assignments:CML:1	v (/op /on) ; pss (V)	expr	$value(vtime("pss "/op") - vtime("pss "/on))$ "Vin" (0.15 0.18 0.21 0.24 0.27 0.3 0.33 0.36 0.39 0.42 0.45 0.48 0.51 0.54 0.57 0.6))	point
Assignments:CML:1	Vop	expr	$vtime("pss "/op \text{ nil})$	point
Assignments:CML:1	Von	expr	$vtime("pss "/on \text{ nil})$	point
Assignments:CML:1	op_peak_swing	expr	$ymax(value(vtime("pss "/op") - vtime("pss "/on))$ "Vin" (0.15 0.18 0.21 0.24 0.27 0.3 0.33 0.36 0.39 0.42 0.45 0.48 0.51 0.54 0.57 0.6)))	point
Assignments:CML:1	Vod	expr	$(vtime("pss "/op \text{ nil}) - vtime("pss "/on \text{ nil}))$	point
Assignments:CML:1	rise time	expr	$riseTime(Vod \ ymin(Vod) \ nil \ ymax(Vod) \ nil \ 10 \ 90 \ nil \ "time")$	point
Assignments:CML:1	Vid	expr	$(vtime("pss "/ip \text{ nil}) - vtime("pss "/in \text{ nil}))$	point
Assignments:CML:1	zero_crossing	expr	$cross(Vod \ 0 \ 1 \ "either" \ nil \ nil)$	point
Assignments:CML:1		expr	$deriv(Vod)$	point
Assignments:CML:1	slew_rate_zero_crossing	expr	$abs(value(deriv(Vod) \ zero_crossing))$	point
Assignments:CML:1	Output Noise (V ² /Hz)	expr	$pow(getData("out" ?result "pnoise_pmjitter") \ 2)$	point
Assignments:CML:1	jitter_PSD	expr	$(calcVal("Output \ Noise \ (V^2/Hz)") / pow(calcVal("slew_rate_zero_crossing") \ 2))$	point
Assignments:CML:1	delay	expr	$delay(?wf1 \ Vid \ ?value1 \ 0 \ ?edge1 \ "rising" \ ?nth1 \ 1 \ ?td1 \ 0.0 \ ?wf2 \ Vod \ ?value2 \ 0 \ ?edge2 \ "rising" \ ?nth2 \ 1 \ ?td2 \ nil \ ?stop \ nil \ ?multiple \ nil)$	point

Figure 28 — Variable Expressions Defined in the Test Bench